

Daily Geometry Drill #3

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions (4/3/2/1 tangents) · Radical Axis by subtraction · Common Chords · Orthogonal Circles ($d^2 = r_1^2 + r_2^2$).

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Compare the centre distance d with $r_1 + r_2$ and $|r_1 - r_2|$, or subtract the equations to read the radical axis.

1. The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.
2. The circles with centre $(0,0)$ radius 5 and centre $(13,0)$ radius 3 are disjoint. Write down the number of common tangents of the two circles.
3. Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y + 1 = 0$ and $x^2 + y^2 - 2x - 4y + 13 = 0$.
4. Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).
5. The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.
6. Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.
7. The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .
8. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .
9. How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?
10. Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Section B Manipulation Drills

(≤50 s each)

Complete the square, locate the centres, then classify the two-circle interaction or compute the radical axis.

1. How many common tangents do the circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x - 6y + 9 = 0$ have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4
2. The radical axis of the circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + 6x - 4y + 10 = 0$ is $2x - y + 3 = 0$. What is $a + b$?
 - A) 3
 - B) -3
 - C) 13
 - D) -13
3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .
 - A) -9
 - B) 9
 - C) -5
 - D) 14
4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.
5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?
 - A) $x = 3$
 - B) $x = 6$
 - C) $y = 3$
 - D) $y = 0$
6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4

7. The circle $x^2 + y^2 = 4$ is orthogonal to $x^2 + y^2 + 8x - 6y + C = 0$. Find C .
- A) 4
B) -4
C) 25
D) 2
8. The circles $x^2 + y^2 = 36$ and $(x - 6)^2 + y^2 = 36$ intersect. Find the length of their common chord.
9. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(4, 8)$. Its possible radii are?
- A) 4 only
B) 20 only
C) 4 and 20
D) 16
10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Section C Substitution & Structure*(≤75 s each)*

Two or three skills at once: classify interactions, subtract for radical axes, and convert radical-axis distances into common chords.

1. The circle with centre $(2, -1)$ and radius 5 and the circle $x^2 + y^2 - 12x - 4y + 24 = 0$ intersect. How do they meet?
- A) at two points
B) at exactly one point (tangent)
C) not at all; one contains the other
D) not at all; disjoint exteriors
2. Two circles of radii 3 and 4 are orthogonal. What is the distance between their centres?
- A) 5
B) 7
C) 1
D) $\sqrt{7}$
3. Two circles of radii 3 and 8 have centres 13 units apart. The direct common tangent segment between the two circles has what length?
- A) 12
B) 5
C) 8

- D) $\sqrt{104}$
4. For the same two circles (radii 3 and 8, centres 13 apart), the transverse common tangent segment has what length?
- A) $4\sqrt{3}$
B) $6\sqrt{3}$
C) 12
D) $2\sqrt{6}$
5. Which of the following is the radical axis of the circles $x^2 + y^2 - 2x - 4y - 8 = 0$ and $x^2 + y^2 + 4x - 2y + 3 = 0$?
- A) $6x + 2y + 11 = 0$
B) $6x - 2y + 11 = 0$
C) $6x + 2y - 11 = 0$
D) $2x + 6y + 11 = 0$
6. How many circles tangent to both coordinate axes pass through the point (9, 2)?
- A) 0
B) 1
C) 2
D) 3
7. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet in two points. What is the length of their common chord?
- A) 12
B) 16
C) 24
D) 26
8. The circles $x^2 + y^2 + 4x - 1 = 0$, $x^2 + y^2 - 8y + 11 = 0$ and $x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes that all pass through a single point. This radical centre has coordinates?
- A) (1, 1)
B) (1, -1)
C) (3, 1)
D) (1, 2)

Section D Challenge Ramp(≤ 90 s each)*Combine radical axes, orthogonality and chord loops end-to-end, then match the options.*

1. A point P moves so that its power with respect to the circle $x^2 + y^2 = 1$ is equal to its power with respect to the circle $(x - 6)^2 + y^2 = 4$. As P moves it traces out?
 - A) the line $x = \frac{11}{4}$
 - B) the circle $x^2 + y^2 = 1$
 - C) a pair of isolated points
 - D) the y -axis
2. The circles $x^2 + y^2 = 100$ and $x^2 + y^2 - 6x - 8y - 40 = 0$ intersect in two points. What is the length of their common chord?
 - A) 8
 - B) 16
 - C) 12
 - D) $6\sqrt{3}$
3. The circle $x^2 + y^2 - 6x - 8y + C = 0$ is orthogonal to the circle with centre $(2, -1)$ and radius 5. Find C .
 - A) 24
 - B) 26
 - C) 25
 - D) -24
4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord?
 - A) $\frac{24}{5}$
 - B) $\frac{12}{5}$
 - C) 6
 - D) $\frac{48}{5}$
5. Three circles of radii 3, 4 and 5 are pairwise externally tangent (each pair has exactly one common point). What is the area of the triangle formed by their three centres?
 - A) $12\sqrt{5}$
 - B) 12
 - C) 84
 - D) $3\sqrt{5}$

“Two circles are decoded by one subtraction: locate each centre, subtract the equations, then compare d with $r_1 + r_2$ and $|r_1 - r_2|$.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths